Indian Institute of Engineering Science and Technology, Shibpur

B. Tech (CST) 5th Semester, Departmental Elective-I, Midterm Examination, 2025

Web Technology (CS3122)

Jagash CSB046

Full Marks: 30 *Time: 2 Hours*

- *Answer all questions*
- *In case of typing mistake/error/confusion in the question paper: Don't panic. Don't call invigilators out. You will get the deserved marks based on an honest attempt.*

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1. **Read the following paragraph as a context, not as the source of all answers.**

Web technology fundamentally relies on the client-server architecture, where a client (like your web browser) requests resources from a server, which then provides them. This interaction is divided into two key areas: Front-end development and Back-end development. Front-end development focuses on everything a user sees and interacts with in their browser, using languages like HTML, CSS, and JavaScript to build the user interface and ensure a seamless, responsive experience. Conversely, Back-end development handles the server-side logic, databases, and APIs that power the application behind the scenes, ensuring data storage, processing, and overall functionality. To manage and deliver these web applications efficiently, cloud computing has become paramount. It provides on-demand access to computing resources (servers, storage, databases) over the internet, eliminating the need for organizations to manage physical infrastructure. Building on this, serverless computing further abstracts away server management, allowing developers to deploy code without provisioning or scaling servers, and only paying for the actual compute time consumed. While cloud computing typically centralizes resources, edge computing pushes computation and data storage closer to the data source or end-user, minimizing latency and bandwidth usage. This distributed approach is particularly beneficial for real-time applications and IoT devices, complementing the centralized nature of cloud and serverless platforms by handling immediate processing at the network's periphery.

A. Choose the correct alternative. No explanation is needed for your answer

(i) Which of the following best describes the role of the client in a client-server architecture?
a) Stores all application data in client computer b) Provides resources to both client and the server c) Requests resources from client to the server d) Handles server-side logic only

(ii) What is the primary focus of Front-end development?
a) Managing databases b) Building a user interface for requesting service c) Handling server-side logic d) Writing code without considering the server

(iii) Which of these technologies aims to minimize latency and bandwidth usage by bringing computation closer to the user or data source?
a) Cloud computing b) Serverless computing c) Back-end development d) Edge computing

(iv) In serverless computing, developers are primarily responsible for:
a) Provisioning and scaling servers b) Managing physical infrastructure c) Deploying code without provisioning or scaling servers d) Designing the user interface

(v) Which of the following is an example of a server-side scripting language?
a) HTML b) JavaScript c) CSS d) PHP

B. Clearly write True or False for the following statements. No justification is needed for your answer.

(i) Back-end development primarily uses HTML, CSS, and JavaScript to build the user interface. F
(ii) Cloud computing eliminates the need for organizations to manage any physical infrastructure. T
(iii) Serverless computing means that there are no servers involved in running the application code. F
(iv) Edge computing is typically beneficial for real-time applications and IoT devices. T
(v) CSS is primarily used for defining the structure and content of a web page. F

C. Match the following. No marks will be awarded for partially correct matching

1. Everything a user sees and interacts with	a. Node.js, PHP, MySQL, Python (2)
2. Server-side logic, databases, and APIs	b. Developers can focus on writing code without worrying about the underlying infrastructure. (4)
3. On-demand access to computing resources over the internet	c. HTML, CSS, JavaScript (1)
4. Platform as a Service	d. Specialized edge gateways/servers. (5)
5. Pushes computation and data storage closer to the data source or end-use	e. Amazon Web Services, Google Cloud Platform, Microsoft Azure (3)

[5+5+5=15]

2. Carefully follow the code

```
<html>
<head><title> My Simple Page</title></head>
<body> <p> I know something better than you </p></body>
</html>
```

Explain the use of <head>, <body> and <p> in this context. Draw the possible output of this code.

[5]

3. Consider a tiny CSS code snippet

```
.my-styled-box {
  background-color: #f0f8ff; /* Alice Blue */
  border: 1px solid #c0d9ec; /* Light blue border */
  padding: 15px;
  margin-bottom: 10px;
  border-radius: 5px;
  font-family: Arial, sans-serif;
  color: #333;
}
```

Given the CSS code snippet above, identify and explain the purpose of three different CSS properties used within the .my-styled-box rule. For each property, describe what visual effect it achieves on an HTML element.

[5]

4. Consider a JavaScript code snippet

```
const simpleButton = document.getElementById('simpleClickMe');
const outputParagraph = document.getElementById('outputMessage');
simpleButton.addEventListener('click', function() {
    outputParagraph.textContent = 'Hello from JavaScript!';
});
```

Explain what do 'document.getElementById()' and 'simpleButton.addEventListener()' do here? Describe the visual outcome when the code is embedded in HTML with <button> and <p>

[5]

Indian Institute of Engineering Science and Technology, Shibpur
BTech (CST) 5th Semester Mid Semester Examination, September, 2025

Computer Architecture and Organization II (CS-3103)

Full Marks: 30 Time: 2 hours

Answer all questions

1. Define predict-taken and predict-not-taken branch prediction schemes. What is an 1-bit predictor? The series of branch outcomes stored in an 1-bit prediction buffer is 11111110 (1 represents taken branch and 0 is for un-taken branch). If the predictor is initialized as 0, then find the number of mis-predictions. 5

2. Define RAW, WAR and WAW hazards. In the pipeline execution of following code sequence

```
SUB R0, R1, R2
MOVE R2, R0,
```

find the type of possible data hazards. 5

3. What is an out-of-order exception in a pipeline system? Explain the solution for handling out-of-order exceptions in a k-stage instruction pipeline with status vectors. Let a status vector register SVi, associated with the instruction Ii, and there can be N types of possible exceptions in the system. Find the minimum number of status vectors required for this system. 5

4. Define "full map directory" and "limited directory" protocols for ensuring cache coherence. The multi-processor systems M1 and M2, each having 2048 processors, implement full map directory and limited directory respectively. The memory block/cache line size for the systems is 128 bytes. The systems allow up to 8 processors to have the cached copy of a block. How many directory state bits are in a sharing vector of M1 and M2 (ignore dirty bit in the directory)? 5

5. B) A bus based shared memory (MM) multiprocessor system consists of three processors P1, P2 and P3 with caches C1, C2, and C3 respectively. Each cache can accommodate *one* memory block at a time. The MM stores *two* blocks B1 and B2. The variables *x and y* are in B1. The variable z is in B2. Initially, in MM $x = 10$, $y = 20$, and $z = 40$. The system follows snoopy based MESI model and write-invalidate scheme. Initially, C1, C2, and C3 are empty. Now, the following events occur in sequence

(1) P1 reads *y*, (2) P2 reads *y*, (3) P2 updates x = *x*-5, (4) P3 reads *y*,
(5) P1 updates y = *y*+2, (6) P3 reads *x*, (7) P2 updates *x* = 10, (8) P1 reads *x*.

i) Show the states of cached blocks in C1, C2 and C3 after each event.
ii) Denote the case of false sharing miss if any. 5

6. A CPU is having 16 KB direct mapped D-cache. The main memory (MM) block size is 64-byte. Suppose A, an 128x128 array, is stored in MM in row major order, where each A[i,j] is of four bytes. The blocks for A are consecutive in MM. Consider the following C code segment

```
for (i = 0; i<128; i++) {
   for (j =0; j<128; j++) {
      y += A[j][i];

 } }
```

a) Find the maximum number of MM blocks that can be accommodated in D-cache at a time.
b) Show whether the loop interchange technique can reduce the cache miss rate. 5

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INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY, SHIBPUR
B.TECH CST (5th sem) Mid-Semester EXAMINATION, 2025

Database Management Systems (CS3102)

F. M. = 30 **Time – 2 hrs.**

Answer all questions

1. i. Given the basic ER and relational models, which of the following is INCORRECT?

(a) An attribute of an entity can have more than one value
(b) An attribute of an entity can be composite
(c) In a row of a relational table, an attribute can have more than one value
(d) In a row of a relational table, an attribute can have exactly one value or a NULL value

ii. Let E1 and E2 be two entities in an E/R diagram with simple single-valued attributes. R1 and R2 are two relationships between E1 and E2, where R1 is one-to-many and R2 is many-to-many. R1 and R2 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model?

(a) 2
(b) 3
(c) 4
(d) 5

iii. The relation book (title, price) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list?

```
select title
from book as B
where (select count(*)
from book as T
where T.price > B.price) < 5
```

(a) Titles of the four most expensive books
(b) Title of the fifth most inexpensive book
(c) Title of the fourth most expensive book
(d) Titles of the five most expensive books

iv. Given the relations employee (name, salary, dept-no), and department (dept-no, dept-name, address)

Which of the following queries cannot be expressed using the basic relational algebra operations ?

(a) Department address of every employee
(b) Employees whose name is the same as their department name
(c) The sum of all employees' salaries
(d) All employees of a given department

v. Consider a B-tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?

a) 1 b) 2 c) 3 d) 4

[5x2]

2. Consider the following relational schema:

Students(rollno: integer, sname: string)
Courses(courseno: integer, cname: string)
Registration(rollno: integer, courseno: integer, percent: real)

Write the query 'Find the distinct names of all students who score more than 90% in the course numbered 3102' in the following form:

a) Relational Algebra
b) Tuple relational calculus
c) SQL

[3+4+3]

3. Draw the ER diagram and design the tables based on the following description:

Banks are identified by a name, code, address of head office. Banks have branches which are identified by a branch_no., branch_name, address. They have customers which are identified by name, cust-id, phone number, address. Customer may have one or more accounts having account_no., acc_type, balance. Customer can avail loans. Loans are identified by loan_id, loan_type and amount. Account and loans are related to bank's branch.

Mention assumptions (if any) during design.

[10]

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B. Tech. (CST) 5th Semester Mid-Term Examination, September, 2025

Microprocessor Based Systems (CS 3101)

Full Marks: 100 (Final Weightage 30) Time: 2 Hours

- **Attempt all questions.**
- **Answers should be precise, to the point, and in your own words as far as practicable.**
- **If not explicitly mentioned, assume Intel 8085A as the base microprocessor.**
- **Make your own assumptions, if necessary, and state them at proper places.**

1. (a) Explain (with suitable examples) the different ways in which parameters and return value can be managed for subroutines in assembly language. What are the pros and cons of each of the approaches.

   (b) Explain the advantage(s) (justification) of having software interrupt instructions in addition to typical subroutine call instructions in the Instruction Set of a microprocessor.

   (c) Write an assembly language subroutine "**unsigned int reverse_digit(int n)**" that reverses the 2 Hex digits of the 8 bit number **n** and returns the reversed number. For example, if **n** is **B5H**, the subroutine should return **5BH**.

   [8+5+7]

2. (a) Say, the **memory space** for an Intel 8085A microprocessor has to be populated with 64Kb (that is, 8K × 8 = 8KB) ROM, 256Kb (that is, 32K × 8 = 32KB) RAM and one Intel 8255 PPIO chip. With a suitable diagram propose and explain how to achieve this.

   (b) Write an assembly language subroutine "**int smallest(int a[ ], unsigned int n)**" that finds and returns the smallest among the **n** elements of the array **a** of 8 bit numbers.

   [12+8]

3. Say, a microprocessor based system is to produce 4 square waves of approximately 10Hz, 20Hz, 30Hz, and 40Hz frequency at 4 of its output lines. Propose a suitable design for the system. Write a suitable assembly language program for the system that will produce the desired output.

   [20]

4. (a) Explain the role of the **Monitor Program** of a typical **Microprocessor based System Development Kit** (eg, SDK-85 kit).

   (b) Given a typical Microprocessor based System Development Kit(**SDK**), **what information** the user must know for utilizing the kit and **why**?

   [10+10]

5. Write short notes on the following.

   (a) IO mapped IO for Intel 8085A microprocessor.

   (b) Memory Folding

   (c) Software versus Hardware Interrupts for Intel 8085A microprocessor

   [6+7+7]

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. (CST) 5th Semester Mid-Semester Examination, September 2025
Graph Algorithms (CS - 3104)

Time: 2 hours **Full Marks: 30**

Answer all questions.

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Do not use any specific programming language to write the pseudo-codes
Write all parts of the same question together.

1. (a) Suppose that a new company has five employees: Zamora, Agraharam, Smith, Chou, and Macintyre. Each employee will assume one of six responsibilities: planning, publicity, sales, marketing, development, and industry relations. Each employee is capable of doing one or more of these jobs: Zamora could do planning, sales, marketing, or industry relations; Agraharam could do planning or development; Smith could do publicity, sales, or industry relations; Chou could do planning, sales, or industry relations; and Macintyre could do planning, publicity, sales, or industry relations.
   i. Model the capabilities of these employees using a bipartite graph.
   ii. Find an assignment of responsibilities such that each employee is assigned one responsibility.
   iii. Is the matching of responsibilities you found in part (ii) a complete matching? Is it a maximum matching?

   (b) Determine whether the given pair of graphs in Fig. 1 is isomorphic by checking all invariants and then using the subgraph matching method.

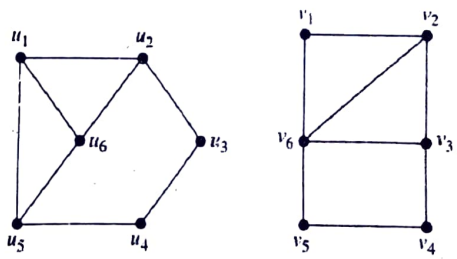


Figure 1: Test for Isomorphism (Graphs G and H)

   (c) What are the properties of a simple graph? Prove the theorem: "An undirected graph has an even number of vertices of odd degree."

   (d) Write pseudo-code for the breadth-first search (BFS) algorithm to generate a spanning tree for a simple graph. Use BFS to find a spanning tree for the graph in Fig. 2 (You may choose a as the root of the tree).

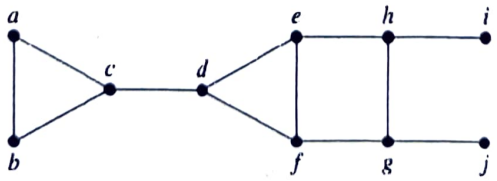


Figure 2: BFS on Graph G

$[(1+1+1)+2+(1+2)+(2+2)=12]$

2. (a) Find $\kappa(G)$, $\lambda(G)$, and $min_{v \in V} deg(v)$ for the graph G shown in Fig. 3. Also, determine which of the two inequalities in $\kappa(G) \leq \lambda(G) \leq min_{v \in V} deg(v)$ are strict.

   (b) Determine whether the two graphs G and H in Fig. 4 are strongly connected and if not, whether it is weakly connected (give brief justifications).

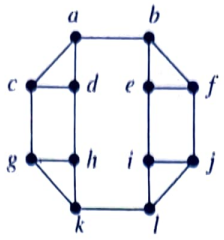


Figure 3: Test of Connectivity for Graph G

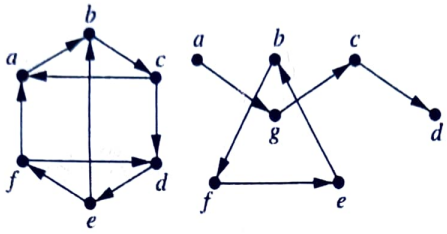


Figure 4: Strongly-connected/Weakly-connected Graphs

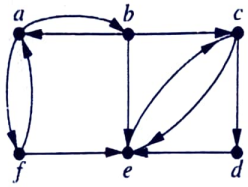


Figure 5: Kosaraju's Algorithm on Graph G

(c) Write a pseudo-code of Kosaraju's algorithm for finding the strongly connected components (SCCs) in a directed graph. Show the step-by-step working of the algorithm to generate all SCCs for the graph G given in Fig. 5.

$$[(1+1+1+1)+(1+1)+(2+2)=10]$$

3. (a) Why does no necessary condition exist for the presence of a Hamilton circuit in a graph?

(b) If a weighted directed graph G contains a negative-weight cycle, can the Bellman-Ford algorithm find the shortest paths if the cycle is reachable from the source? Explain with an example.

(c) Write pseudo-code for Dijkstra's algorithm to find single-source shortest paths in a given simple graph. Show the step-by-step working of the algorithm in finding the shortest paths from source s in the graph shown in Fig. 6.

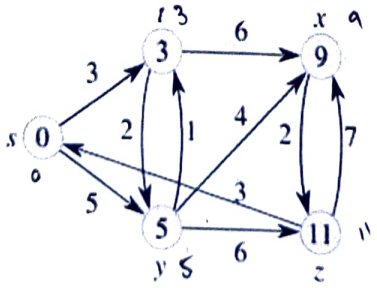


Figure 6: Dijkstra's Algorithm on Graph G

$$[1+2+(2+3)=8]$$